

Figure 2-7. *Compiler error*

This can be a little disappointing at first, but hang in there, I'm sure you'll figure it out. There are two general types of errors that we should be concerned with: hardware and syntax. Coming up are steps for troubleshooting these various error codes to help you get going again.

General **hardware errors** occur because something does not match the hardware you have or a connection isn't made correctly, as in the following example errors:

```
Serial port not found.

Problem uploading to board.
```

First, check that your board is connected to your computer and is indicating good power with a lit LED marked ON. With this verified, next check that the correct serial port and the proper board type have both been selected under the Tools menu. If the USB cable is not connected or the wrong serial port has been selected, the development environment will offer to connect using a different serial port. Sometimes a more serious hardware failure will generate similar errors and you will need to talk to your reseller to iron these problems out.

Other types of hardware errors might show up as something not working the way you think it should because of a bad connection. In our example project, you would expect the LED to first blink red, then green, and then blue; but yours doesn't light up red for some reason. This type of error comes from not wiring the components correctly, reversing the direction of certain parts like LEDs, or having the wires go to the wrong pins. This can be easily fixed by carefully going back over the hardware circuit, as shown in the schematic and illustration, to find out where something went wrong. It's even a good idea to have another pair of eyes look over it just in case they see something that you missed.

The second category of common errors are **syntactical errors** where you missed a semicolon, parenthesis, curly brace, or other character or form of syntax needed for the sketch to function properly, as shown in Figure 2-7 and the following examples: